Positive solutions of the Felder–Hoppe recurrences for quantum minimal surfaces

Leslie P. Polzer

Independent Researcher

polzer@fastmail.com

October 23, 2026

Abstract. For every integer $d \geq 2$ and every $a > 0$, we prove the conjecture of Felder and Hoppe that the leading Gram matrices of their indefinite Hermitian form with density $\exp\bigl(-a|z|^2 + \frac{\mathrm{i}}{d}(z^d + \bar{z}^d)\bigr)$ are nondegenerate, and we show that all signed norms of the corresponding orthogonal polynomials are positive. Consequently, the discrete Painlevé-type recurrence satisfied by the norm ratios, which arises from the quantum minimal surface $w_2 = w_1^{d-1}$, has a positive solution with the moment initial data of Felder and Hoppe. The proof writes the planar integral that generates each symmetry block as the Laplace transform of a product of $d-1$ independent Gamma variables. The Gram blocks become Hankel matrices of these Laplace transforms, and total positivity of the logarithmic Gamma kernels, together with Andréief's identity, fixes the sign of every leading minor. For $d = 4$ we identify the initial data with Bessel functions of orders $\pm\frac{1}{4}$, and we show, by a weighted energy estimate for the convex variational structure of the recurrence, that the positive solution is unique.

Keywords. discrete Painlevé-type recurrence; indefinite orthogonal polynomials; total positivity; Gamma distribution; quantum minimal surface.

MSC2020 Classification. 34M55; 39A36; 33C45; 15B48; 53A10.

1. Introduction

Discrete Painlevé-type recurrences occur both in matrix regularizations of minimal surfaces and as recurrences for the coefficients of orthogonal polynomials. For the quantum minimal surfaces associated with the holomorphic curves $w_2 = w_1^{d-1}$, recurrences of this type were obtained in [CT99, AHK19, Hop25] (see [FH25, Section 3.1]). The quartic recurrence is written explicitly in [Hop25, equation (38)] and [CDHM25, Section 5]; those works also treat the positive special-function solution in the cubic case $d = 3$.

Felder and Hoppe [FH25] constructed an indefinite Hermitian form for each d (recalled in Section 2) and showed that, whenever its leading Gram matrices are nondegenerate, the ratios V_n of its orthogonal-polynomial norms obey

$$V_n + \frac{1}{a^2} \sum_{j=0}^{d-2} \prod_{k=0}^{d-2} V_{n+j-k} = \frac{n+1}{a}, \qquad n \geq 0, \tag{1.1}$$

where $V_n = 0$ for $n < 0$, with the initial values

$$V_j = \frac{\int_{\mathbb{C}} |z|^{2j+2} \rho_a \, \mathrm{d}A}{\int_{\mathbb{C}} |z|^{2j} \rho_a \, \mathrm{d}A}, \qquad 0 \le j \le d-3, \tag{1.2}$$

where ρ_a is the density (2.1) [FH25, equation (3.3) and the display after it]. They proved the required Gram nondegeneracy for $d = 1, 2, 3$ and conjectured it for every d [FH25, Conjecture 3.6 and Theorem 3.7]. Their conjecture is stated for $a \ge 0$. We treat $a > 0$, where all planar moments converge absolutely; at $a = 0$ they do not, and the endpoint would require an oscillatory or regularized interpretation, which we do not discuss.

Theorem 1.1. *Let $d \ge 2$ and $a > 0$.*

(i) *Every leading Gram matrix of the Felder–Hoppe Hermitian form* (Section 2) *is nondegenerate, and all its signed norms $\mathfrak{h}_n$ are positive. Thus [FH25, Conjecture 3.6] holds for every d and every $a > 0$.*

(ii) *Let $d \ge 3$. The ratios $V_n = \mathfrak{h}_{n+1}/\mathfrak{h}_n$ form the unique real sequence that satisfies* (1.1) *for every $n \ge 0$ and has the initial values* (1.2)*. In particular, this solution is positive.*

The moments in (1.2) are explicit expectations of Gamma variables (Proposition 4.1). For $d = 4$, after setting $v_n = V_n/a$ and $\epsilon = a^{-2}$, the recurrence (1.1) becomes

$$v_n \left(1 + v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}\right) = (n+1)\epsilon, \qquad n \ge 0, \tag{1.3}$$

where $v_n = 0$ for $n < 0$. The two initial values are

$$v_0 = -\frac{1}{a}\frac{h'(a)}{h(a)}, \qquad v_1 = -\frac{1}{a}\frac{h''(a)}{h'(a)}, \tag{1.4}$$

with

$$\begin{aligned} h(a) = \frac{\pi^2 a}{4}\Bigg[& J_{-1/4}\left(\frac{a^2}{4}\right)^2 - \sqrt{2}\, J_{-1/4}\left(\frac{a^2}{4}\right) J_{1/4}\left(\frac{a^2}{4}\right) \\ & + J_{1/4}\left(\frac{a^2}{4}\right)^2 \Bigg]. \end{aligned} \tag{1.5}$$

This is the function $h(a) = \int_{\mathbb{C}} \rho_a \, \mathrm{d}A$ of [FH25, equation (3.4)] for $d = 4$, in the Bessel form given in [FH25, Section 4]; Lemma 7.2 below rederives it.

Theorem 1.2. *Let $a > 0$. The initial values* (1.4)–(1.5) *are well defined and extend uniquely to a real sequence satisfying* (1.3) *for every $n \ge 0$. This sequence is $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$ for $d = 4$; in particular, $v_n(a) > 0$ for every $n \ge 0$.*

Uniqueness in Theorem 1.2 refers to the prescribed initial data. The next theorem shows that the positive solution itself is unique, so that the Bessel values (1.4) are the only initial values leading to a positive solution.

Theorem 1.3. *For every $\epsilon > 0$, there is at most one sequence of positive numbers $(v_n)_{n\geq 0}$ satisfying (1.3) for every $n \geq 0$. In particular, for $a > 0$ and $\epsilon = a^{-2}$, the sequence of Theorem 1.2 is the unique positive solution of (1.3).*

For the cubic analogue, the existence of a positive solution was proved in [AHK19, Section 7] and its uniqueness in [CDHM25, Theorem 2.1, proved in Section 4.2]; for the quartic recurrence, [CDHM25, Section 5] reports preliminary evidence for a positive solution with analogous properties. Our proof of Theorem 1.3 (Section 8) uses only the recurrence and is independent of the rest of the paper: in logarithmic variables the recurrence is the stationarity condition of a convex functional, and a weighted energy estimate controls the boundary at infinity. For $d \geq 5$ its a priori bounds are too weak (Remark 8.1), and we do not know whether the positive solution is unique.

The proof of Theorem 1.1(i) treats all d and all indices at once. Independent source parameters u and v for z^d and $\bar{z}^d$ are used on disks, where the two-source integral is entire and, by rotation invariance, depends only on uv. In exponential coordinates $u = \mathrm{e}^{t_1}$, $v = \mathrm{e}^{t_2}$ this turns the Gram block of each residue class modulo d, after a unitriangular Stirling transformation, into the Hankel matrix of derivatives of a single kernel $F_r(t)$, the planar integral with weight $|z|^{2r}$ and coupling $\mathrm{e}^{t/2}$ (Lemma 3.1). A Mellin transform and Gauss's multiplication formula show that F_r is the Laplace transform, in e^t, of a product of $d-1$ independent Gamma variables (Proposition 4.1). The signs of its Hankel determinants then follow from total positivity: the densities of logarithms of Gamma variables are totally positive convolution kernels, and Andréief's identity writes each Hankel determinant as an integral of a product of two determinants of fixed sign (Lemma 5.1). This determinant step is a continuous Cauchy–Binet argument of the kind used for biorthogonal ensembles [Bor99]. To our knowledge, the new ingredients are the two-source representation, which separates the insertions of z^d and $\bar{z}^d$ that a common coupling mixes, and the Gamma-product form of the kernels.

Section 2 recalls the Hermitian form. Section 3 reduces the Gram blocks to Hankel determinants, Section 4 proves the Gamma representation, and Section 5 proves Theorem 1.1(i). Section 6 derives the recurrence and proves Theorem 1.1(ii). Section 7 treats $d = 4$ and proves Theorem 1.2, and Section 8 proves Theorem 1.3. For $d = 4$, Theorem 1.1 together with Theorems 1.2 and 1.3 has been formally verified in Lean 4.33.0 with Mathlib, as the declarations `Painleve.main_theorem` and `Painleve.unique_theorem`. That formalization follows an earlier, quartic-specific form of the sign argument (see the code availability statement); the general-d argument of Sections 3–6 has not been formalized.

2. The Gram form

Fix $d \geq 2$ and $a > 0$, and put

$$\rho_a(z) = \exp\left(-a|z|^2 + \frac{\mathrm{i}}{d}(z^d + \bar{z}^d)\right), \tag{2.1}$$

the density of [FH25, Section 3.1]. The Gaussian factor makes all polynomial moments absolutely convergent. Let $\eta = \mathrm{e}^{\pi \mathrm{i}/d}$. Since $\eta^d = -1$,

$$\rho_a(\zeta z) = \rho_a(z) \quad (\zeta^d = 1), \qquad \rho_a(\eta z) = \overline{\rho_a(z)}.$$

Decompose the polynomial ring into weight spaces

$$\mathbb{C}[z] = \bigoplus_{r=0}^{d-1} \mathcal{P}_r, \qquad \mathcal{P}_r = \{f \in \mathbb{C}[z] : f(\zeta z) = \zeta^r f(z) \text{ for } \zeta^d = 1\} = \operatorname{span}\{z^{r+dq} : q \geq 0\}.$$

For $f \in \mathcal{P}_r$, set $(\sigma f)(z) = (-\eta)^{-r} f(\eta z)$, extend σ linearly to $\mathbb{C}[z]$, and define

$$(f, g)_a = \int_{\mathbb{C}} f(z)\overline{\sigma g(z)}\rho_a(z)\,\mathrm{d}A(z), \qquad \mathrm{d}A = \mathrm{d}(\operatorname{Re} z)\,\mathrm{d}(\operatorname{Im} z), \tag{2.2}$$

which is linear in f and conjugate-linear in g. This is the Hermitian form of [FH25, Proposition 3.1], with the right inverse $\chi(r) = r$ $(0 \leq r \leq d-1)$ chosen there. If $m = r + dq$ with $0 \leq r \leq d-1$, then

$$\sigma z^m = (-\eta)^{-r}\eta^m z^m = (-1)^{r+q} z^m. \tag{2.3}$$

Since σ preserves each $\mathcal{P}_r$ and ρ_a is invariant under $z \mapsto \zeta z$, distinct weight spaces are orthogonal. On $\mathcal{P}_r$, in the basis $z^r, z^{r+d}, \ldots$, the Gram matrix is therefore

$$G^{(r)}_{pq} = (z^{r+dp}, z^{r+dq})_a = (-1)^{r+q} M^{(r)}_{pq}, \qquad M^{(r)}_{pq} = \int_{\mathbb{C}} z^{r+dp}\bar{z}^{r+dq}\rho_a(z)\,\mathrm{d}A(z). \tag{2.4}$$

The substitution $z \mapsto \eta z$ and $\rho_a(\eta z) = \overline{\rho_a(z)}$ give $\overline{M^{(r)}_{qp}} = (-1)^{q-p} M^{(r)}_{pq}$, so each $G^{(r)}$ is a Hermitian matrix.

For $k \geq 1$, let

$$\Delta^{(r)}_k = \det[G^{(r)}_{pq}]_{p,q=0}^{k-1}, \qquad \Delta^{(r)}_0 = 1. \tag{2.5}$$

Assume for the moment that all $\Delta^{(r)}_k$ are nonzero. After the monomials $1, z, \ldots, z^N$ are grouped by residue modulo d, their Gram matrix is block diagonal, with leading principal submatrices of the $G^{(r)}$ as blocks. Hence every leading Gram matrix $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and Gram–Schmidt gives unique monic orthogonal polynomials $P_n(z) = z^n + \cdots$. Orthogonalizing z^n against lower monomials subtracts only components in the weight space of z^n, so $P_n \in \mathcal{P}_{n \bmod d}$. Following [FH25, Lemma 3.4 and Example 3.5], for $\ell \in \mathbb{Z}$ put

$$\tau(\ell) = \begin{cases} (-1)^{d-1}, & \ell \equiv -1 \pmod d, \\ 1, & \text{otherwise}, \end{cases} \qquad \varrho(n) = (-1)^n \prod_{\ell=0}^{n-1} \tau(\ell). \tag{2.6}$$

If $n = r + dj$ with $0 \leq r \leq d-1$, exactly j of the integers $0, \ldots, n-1$ are congruent to $d-1$, so

$$\varrho(r + dj) = (-1)^{r+dj+(d-1)j} = (-1)^{r+j}. \tag{2.7}$$

The signed norms $\mathfrak{h}_n$ of [FH25, Section 3.1], denoted h_n there, are defined by

$$(P_n, P_n)_a = \varrho(n)\mathfrak{h}_n. \tag{2.8}$$

The matrices $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ are exactly the Gram matrices of [FH25, Conjecture 3.6].

3. Gram blocks as Hankel determinants

We write $\binom{k}{2} = k(k-1)/2$ and $\left\{ {q \atop j} \right\}$ for the Stirling numbers of the second kind, defined by $\left\{ {0 \atop 0} \right\} = 1$, $\left\{ {q \atop j} \right\} = 0$ for $j > q$ or $j = 0 < q$, and $\left\{ {q+1 \atop j} \right\} = j\left\{ {q \atop j} \right\} + \left\{ {q \atop j-1} \right\}$. For a function G that is smooth on $(0,\infty)$, induction on q gives

$$\frac{\mathrm{d}^q}{\mathrm{d}t^q} G(\mathrm{e}^t) = \sum_{j=0}^{q} \left\{ {q \atop j} \right\} \mathrm{e}^{jt} G^{(j)}(\mathrm{e}^t), \tag{3.1}$$

because $\frac{\mathrm{d}}{\mathrm{d}t}\left[\mathrm{e}^{jt}G^{(j)}(\mathrm{e}^t)\right] = j\mathrm{e}^{jt}G^{(j)}(\mathrm{e}^t) + \mathrm{e}^{(j+1)t}G^{(j+1)}(\mathrm{e}^t)$.

For $r \in \{0,\dots,d-1\}$ and $\lambda \geq 0$, define the weighted sources

$$\mathcal{W}_r(a;\lambda) = \int_{\mathbb{C}} |z|^{2r} \exp\left(-a|z|^2 + \frac{\mathrm{i}\lambda}{d}(z^d + \bar{z}^d)\right) \mathrm{d}A(z), \qquad F_r(t) = \mathcal{W}_r(a;\mathrm{e}^{t/2}).$$

Lemma 3.1 *(two-source identity).* *For $r \in \{0,\dots,d-1\}$ and all $p,q \geq 0$,*

$$\sum_{k=0}^{p}\sum_{l=0}^{q} \left\{ {p \atop k} \right\}\left\{ {q \atop l} \right\} \left(\frac{\mathrm{i}}{d}\right)^{k+l} M_{kl}^{(r)} = F_r^{(p+q)}(0). \tag{3.2}$$

Proof. For $R > 0$ and $u,v \in \mathbb{C}$, define the disk integral

$$\mathcal{Z}_R(u,v) = \int_{|z|\leq R} |z|^{2r} \exp\left(-a|z|^2 + \frac{\mathrm{i}}{d}(uz^d + v\bar{z}^d)\right) \mathrm{d}A(z).$$

Over the compact disk, all derivatives in (u,v) may be taken under the integral sign, and $\mathcal{Z}_R$ is entire in each variable.

Step 1: $\mathcal{Z}_R(u,v) = \mathcal{Z}_R(uv,1)$ for $u,v > 0$. For fixed $u,v > 0$, the function $H(s) = \mathcal{Z}_R(u\mathrm{e}^s, v\mathrm{e}^{-s})$ is entire in $s \in \mathbb{C}$. The rotation $z \mapsto \mathrm{e}^{\mathrm{i}\varphi/d}z$, $\varphi \in \mathbb{R}$, preserves the disk, $|z|$ and the area measure, and multiplies z^d by $\mathrm{e}^{\mathrm{i}\varphi}$ and $\bar{z}^d$ by $\mathrm{e}^{-\mathrm{i}\varphi}$; hence $H(s+\mathrm{i}\varphi) = H(s)$. Differentiating in φ at $\varphi = 0$ gives $\mathrm{i}H'(s) = 0$, so H is constant, and $H(0) = H(\log v)$ is the claim.

Step 2: the truncated identity. For real t_1, t_2 put $K(t_1,t_2) = \mathcal{Z}_R(\mathrm{e}^{t_1},\mathrm{e}^{t_2})$. By Step 1, $K(t_1,t_2) = f_R(t_1+t_2)$ with $f_R(t) = \mathcal{Z}_R(\mathrm{e}^{t/2},\mathrm{e}^{t/2})$, so $\partial_{t_2}^q\partial_{t_1}^p K(0,0) = f_R^{(p+q)}(0)$. On the other hand, (3.1) with $G(w) = \exp(cw)$ gives $\frac{\mathrm{d}^p}{\mathrm{d}t^p}\exp(c\mathrm{e}^t) = \sum_k \left\{ {p \atop k} \right\}(c\mathrm{e}^t)^k \exp(c\mathrm{e}^t)$; applying this under the integral, first with $c = \mathrm{i}z^d/d$ and then with $c = \mathrm{i}\bar{z}^d/d$, yields

$$\partial_{t_2}^q\partial_{t_1}^p K(0,0) = \sum_{k=0}^{p}\sum_{l=0}^{q} \left\{ {p \atop k} \right\}\left\{ {q \atop l} \right\} \left(\frac{\mathrm{i}}{d}\right)^{k+l} \int_{|z|\leq R} z^{r+dk}\bar{z}^{r+dl}\rho_a(z)\,\mathrm{d}A(z),$$

where $|z|^{2r}z^{dk}\bar{z}^{dl} = z^{r+dk}\bar{z}^{r+dl}$.

Step 3: $R \to \infty$. The truncated moments converge to $M_{kl}^{(r)}$, since their integrands are bounded by $|z|^{2r+dk+dl}\mathrm{e}^{-a|z|^2}$. For real t, the factor $\exp\left(\frac{\mathrm{i}}{d}\mathrm{e}^{t/2}(z^d+\bar{z}^d)\right)$ has modulus 1, and by (3.1) its t-derivatives of order m are this factor times a polynomial in $\mathrm{e}^{t/2}(z^d+\bar{z}^d)$; for $|t| \leq 1$ they are bounded by $c_m(1+|z|)^{dm}$. Hence derivatives of all orders may be taken under the integral both for f_R and for F_r, and dominated convergence gives $f_R^{(m)}(0) \to F_r^{(m)}(0)$. Letting $R \to \infty$ in Step 2 proves (3.2). ∎

Corollary 3.2. *For $r \in \{0,\dots,d-1\}$ and $k \geq 1$,*

$$\Delta_k^{(r)} \;=\; (-1)^{rk} d^{2\binom{k}{2}} \det\Big[F_r^{(p+q)}(0)\Big]_{p,q=0}^{k-1}. \tag{3.3}$$

Proof. By (2.4), $G_{pq}^{(r)} = (-1)^{r+q}(d/\mathrm{i})^{p+q}\big[(\mathrm{i}/d)^{p+q} M_{pq}^{(r)}\big]$. Taking $(d/\mathrm{i})^p$ out of row p and $(-1)^{r+q}(d/\mathrm{i})^q$ out of column q, and using $(d/\mathrm{i})^2 = -d^2$, gives $\Delta_k^{(r)} = (-1)^{rk} d^{2\binom{k}{2}} \det[(\mathrm{i}/d)^{p+q} M_{pq}^{(r)}]_{p,q<k}$. The left-hand side of (3.2) is the (p,q) entry of $\Sigma\,[(\mathrm{i}/d)^{k+l} M_{kl}^{(r)}]\,\Sigma^{\mathsf{T}}$, where $\Sigma_{pk} = \left\{{p \atop k}\right\}$ is lower triangular with ones on the diagonal; its $k \times k$ truncations therefore have the same determinant. ∎

4. The kernels as Laplace transforms

A random variable G has the Gamma distribution with shape $\beta > 0$ if it has density $x^{\beta-1}\mathrm{e}^{-x}/\Gamma(\beta)$ on $(0,\infty)$; then $\mathbb{E}[G^{-z}] = \Gamma(\beta - z)/\Gamma(\beta)$ for $\operatorname{Re} z < \beta$. For $r \in \{0,\dots,d-1\}$ let

$$J_r \;=\; \{0,1,\dots,d-1\} \setminus \{d-1-r\}, \qquad \beta_{r,j} \;=\; \frac{r+1+j}{d} \quad (j \in J_r),$$

let $G_{r,j}$, $j \in J_r$, be independent Gamma variables with shapes $\beta_{r,j}$, and put $S_r = \prod_{j \in J_r} G_{r,j}$, a product of $d-1$ factors.

Proposition 4.1 *(Gamma representation).* *For $r \in \{0,\dots,d-1\}$ and $\lambda \geq 0$,*

$$\mathcal{W}_r(a;\lambda) \;=\; \pi\, r!\, a^{-r-1}\, \mathbb{E}\left[\exp\!\big(-\lambda^2 d^{\,d-2} a^{-d} S_r\big)\right]. \tag{4.1}$$

In particular, $\int_{\mathbb{C}} |z|^{2r} \rho_a \,\mathrm{d}A = \pi r!\, a^{-r-1}\mathbb{E}[\exp(-d^{\,d-2}a^{-d}S_r)]$, and $F_r(t) = \pi r!\, a^{-r-1}\mathbb{E}[\exp(-\mathrm{e}^t \kappa S_r)]$ with $\kappa = d^{\,d-2}a^{-d}$.

Proof. At $\lambda = 0$ both sides equal $\pi r!\, a^{-r-1}$. The substitution $z \mapsto a^{-1/2} z$ gives $\mathcal{W}_r(a;\lambda) = a^{-r-1}\mathcal{W}_r(1;\lambda a^{-d/2})$, so we may take $a = 1$ and $\lambda > 0$.

Step 1: a radial integral. In polar coordinates $z = R\mathrm{e}^{\mathrm{i}\theta}$, $z^d + \bar{z}^d = 2R^d\cos(d\theta)$, and $\int_0^{2\pi} \mathrm{e}^{\mathrm{i}x\cos(d\theta)}\,\mathrm{d}\theta = 2\pi J_0(x)$ [OLBC10, equation (10.9.2)]. With $u = R^2$,

$$\mathcal{W}_r(1;\lambda) \;=\; \pi \int_0^\infty u^r \mathrm{e}^{-u} J_0\big(c\lambda u^{d/2}\big)\,\mathrm{d}u, \qquad c \;=\; \frac{2}{d}.$$

Step 2: the Mellin transform. Write $g(x) = \mathcal{W}_r(1;\sqrt{x})$ for $x > 0$, and let $0 < \operatorname{Re} z = \sigma < \min(\frac14, \frac{r+1}{d})$. Since $|J_0(y)| \leq 1$ and $|J_0(y)| \leq Cy^{-1/2}$ [OLBC10, equations (10.14.1) and (10.17.3)], $\int_0^\infty y^{2\sigma-1}|J_0(y)|\,\mathrm{d}y < \infty$. The substitution $y = c\sqrt{x}\,u^{d/2}$ gives $\int_0^\infty x^{\sigma-1}|J_0(c\sqrt{x}\,u^{d/2})|\,\mathrm{d}x = 2(cu^{d/2})^{-2\sigma}\int_0^\infty y^{2\sigma-1}|J_0(y)|\,\mathrm{d}y$, and $\int_0^\infty u^{r-d\sigma}\mathrm{e}^{-u}\,\mathrm{d}u < \infty$. Hence Fubini's theorem applies, and with $\int_0^\infty y^{2z-1}J_0(y)\,\mathrm{d}y = 2^{2z-1}\Gamma(z)/\Gamma(1-z)$ [OLBC10, equation (10.22.43)] we obtain

$$\int_0^\infty x^{z-1} g(x)\,\mathrm{d}x \;=\; \pi\, d^{2z}\, \frac{\Gamma(z)\,\Gamma(r+1-dz)}{\Gamma(1-z)}.$$

Step 3: Gauss's multiplication formula. By [OLBC10, equation (5.5.6)], applied at $r+1-dz$ and at $r+1$,

$$\frac{\Gamma(r+1-dz)}{\Gamma(r+1)} = d^{-dz}\prod_{j=0}^{d-1}\frac{\Gamma\left(\frac{r+1+j}{d}-z\right)}{\Gamma\left(\frac{r+1+j}{d}\right)}.$$

The factor with $j = d-1-r$ equals $\Gamma(1-z)$. Therefore

$$\int_0^\infty x^{z-1}g(x)\,\mathrm{d}x = \pi\, r!\,\Gamma(z)\left(d^{\,d-2}\right)^{-z}\prod_{j\in J_r}\frac{\Gamma(\beta_{r,j}-z)}{\Gamma(\beta_{r,j})} = \pi\, r!\,\Gamma(z)\,\mathbb{E}\big[(d^{\,d-2}S_r)^{-z}\big].$$

Step 4: identification. Let $\tilde g(x) = \pi r!\,\mathbb{E}[\exp(-x\,d^{\,d-2}S_r)]$. By Tonelli's theorem and $\int_0^\infty x^{z-1}\mathrm{e}^{-xs}\,\mathrm{d}x = \Gamma(z)s^{-z}$, the Mellin transform of $\tilde g$ is the last expression for $0 < \operatorname{Re} z < \min_{j\in J_r}\beta_{r,j}$, a strip that contains the line $\operatorname{Re} z = \sigma$ of Step 2. Both g and $\tilde g$ are continuous on $(0,\infty)$, and $x^{\sigma-1}g$, $x^{\sigma-1}\tilde g$ are integrable. Under $x = \mathrm{e}^s$, their Mellin transforms on $\operatorname{Re} z = \sigma$ are the Fourier transforms of the integrable functions $\mathrm{e}^{\sigma s}g(\mathrm{e}^s)$ and $\mathrm{e}^{\sigma s}\tilde g(\mathrm{e}^s)$; these coincide, so $g = \tilde g$. ∎

5. Signs of the Hankel determinants

For real $x_1,\dots,x_k$ write $V(x_1,\dots,x_k) = \prod_{1\le i<j\le k}(x_j-x_i)$.

Lemma 5.1 *(Hankel signs).* *Let $m \ge 1$, $\beta_1,\dots,\beta_m > 0$ and $\kappa > 0$, let $G_1,\dots,G_m$ be independent Gamma variables with shapes $\beta_1,\dots,\beta_m$, and let $F(t) = \mathbb{E}[\exp(-\mathrm{e}^t\kappa G_1\cdots G_m)]$. Then, for every real t and every $k \ge 1$,*

$$(-1)^{\binom{k}{2}}\det\left[F^{(p+q)}(t)\right]_{p,q=0}^{k-1} > 0.$$

Proof. The variable $\log G_j$ has the density $q_j(v) = \mathrm{e}^{\beta_j v-\mathrm{e}^v}/\Gamma(\beta_j)$ on $\mathbb{R}$. Put $\phi_0(u) = \exp(-\mathrm{e}^u)$ and, for $j = 1,\dots,m$,

$$\phi_j(u) = \mathbb{E}\big[\phi_{j-1}(u+\log G_j)\big] = \int_{\mathbb{R}}\phi_{j-1}(w)\,q_j(w-u)\,\mathrm{d}w,$$

so that $F(t) = \phi_m(t+\log\kappa)$. By (3.1) with $G(x) = \mathrm{e}^{-x}$, $\phi_0^{(p)}(u) = \mathrm{e}^{-x}\sum_i \left\{{p \atop i}\right\}(-x)^i$ with $x = \mathrm{e}^u$; in particular $\phi_0^{(p)}$ is bounded. By induction and dominated convergence, each ϕ_j is smooth with bounded derivatives $\phi_j^{(p)}(u) = \mathbb{E}[\phi_{j-1}^{(p)}(u+\log G_j)] = \int\phi_{j-1}^{(p)}(w)q_j(w-u)\,\mathrm{d}w$.

Claim: for $0 \le j \le m-1$ and $u_1 < \cdots < u_k$, $\operatorname{sgn}\det[\phi_j^{(p)}(u_l)]_{p<k,\,l\le k} = (-1)^{\binom{k}{2}}$. For $j = 0$, the matrix Σ of Corollary 3.2 is unitriangular, so with $x_l = \mathrm{e}^{u_l}$,

$$\det[\phi_0^{(p)}(u_l)] = \mathrm{e}^{-\sum_l x_l}V(-x_1,\dots,-x_k) = (-1)^{\binom{k}{2}}\mathrm{e}^{-\sum_l x_l}V(x_1,\dots,x_k),$$

and $V(x) > 0$ because the x_l increase. For the induction step, Andréief's identity [And86, For19], applied to the bounded functions $\phi_{j-1}^{(p)}$ and the integrable functions $w \mapsto q_j(w-u_l)$, together with the symmetry of the integrand, gives

$$\det[\phi_j^{(p)}(u_l)] = \int_{w_1<\cdots<w_k}\det[\phi_{j-1}^{(p)}(w_i)]_{p,i}\,\det[q_j(w_i-u_l)]_{i,l}\prod_i\mathrm{d}w_i.$$

Here $q_j(w-u) = \Gamma(\beta_j)^{-1}\mathrm{e}^{\beta_j w}\mathrm{e}^{-\beta_j u}\exp(\xi\zeta)$ with $\xi = \mathrm{e}^w$ and $\zeta = -\mathrm{e}^{-u}$, both strictly increasing. The kernel $\exp(\xi\zeta)$ is strictly totally positive: for $\xi_1 < \cdots < \xi_k$ and $\zeta_1 < \cdots < \zeta_k$, $\det[\mathrm{e}^{\xi_i\zeta_l}] > 0$. Indeed, a nonzero sum $\sum_l c_l \mathrm{e}^{\zeta_l \xi}$ has at most $k-1$ real zeros, by induction on k with Rolle's theorem applied to $\frac{\mathrm{d}}{\mathrm{d}\xi}(\mathrm{e}^{-\zeta_1\xi}\sum_l c_l \mathrm{e}^{\zeta_l\xi})$; so the determinant never vanishes on the connected set of ordered (ξ,ζ). To find its sign, fix $x_1 < \cdots < x_k$ and put $\xi_i = \varepsilon x_i$. Expanding $\mathrm{e}^{\varepsilon x_i\zeta_l} = \sum_{p\ge 0}(\varepsilon x_i\zeta_l)^p/p!$ and applying the Cauchy–Binet formula gives

$$\det[\mathrm{e}^{\varepsilon x_i\zeta_l}] = \varepsilon^{\binom{k}{2}}\,\frac{V(x_1,\dots,x_k)\,V(\zeta_1,\dots,\zeta_k)}{\prod_{p=0}^{k-1} p!} + O\big(\varepsilon^{\binom{k}{2}+1}\big) \qquad (\varepsilon \downarrow 0),$$

which is positive for small ε. Hence the second determinant is positive on the ordered region, the first has sign $(-1)^{\binom{k}{2}}$ by induction, and the claim follows.

Now let $u = t + \log\kappa$ and $H_{pq} = \phi_m^{(p+q)}(u)$. From $q_m' = q_m\cdot(\beta_m - \mathrm{e}^v)$, induction gives $q_m^{(q)}(v) = q_m(v)P_q(\mathrm{e}^v)$, where $P_0 = 1$ and $P_{q+1}(y) = (\beta_m - y)P_q(y) + yP_q'(y)$, so that P_q has degree q and leading coefficient $(-1)^q$. On $|u|\le M$, $|q_m^{(q)}(w-u)| \le C_{q,M}\mathrm{e}^{\beta_m w}(1+\mathrm{e}^w)^q\exp(-\mathrm{e}^{-M}\mathrm{e}^w)$, which is integrable in w. Differentiating $\phi_m^{(p)}(u) = \int \phi_{m-1}^{(p)}(w)q_m(w-u)\,\mathrm{d}w$ a further q times in u therefore gives

$$H_{pq} = \int_{\mathbb{R}} \phi_{m-1}^{(p)}(w)\,g_q(w)\,\mathrm{d}w, \qquad g_q(w) = (-1)^q q_m^{(q)}(w-u) = q_m(w-u)\,\widetilde{P}_q(\mathrm{e}^{w-u}),$$

with $\widetilde{P}_q = (-1)^q P_q$ monic of degree q. Andréief's identity gives

$$\det H = \int_{w_1<\cdots<w_k} \det[\phi_{m-1}^{(p)}(w_i)]\prod_i q_m(w_i-u)\;V\big(\mathrm{e}^{w_1-u},\dots,\mathrm{e}^{w_k-u}\big)\prod_i \mathrm{d}w_i,$$

because $\det[\widetilde{P}_q(y_i)] = V(y)$ for monic $\widetilde{P}_q$. On the ordered region $V(\mathrm{e}^{w_i-u}) > 0$, and by the claim the integrand has sign $(-1)^{\binom{k}{2}}$ and does not vanish. Hence $(-1)^{\binom{k}{2}}\det H > 0$. ∎

Proposition 5.2 *(block determinants).* *For $r\in\{0,\dots,d-1\}$ and $k\ge 1$, $\operatorname{sgn}\Delta_k^{(r)} = (-1)^{rk+\binom{k}{2}}$.*

Proof. By Proposition 4.1, $F_r(t) = \pi r!\,a^{-r-1}F(t)$, where F is the function of Lemma 5.1 for the $m = d-1 \ge 1$ shapes $\beta_{r,j}$, $j\in J_r$, and $\kappa = d^{\,d-2}a^{-d}$. Hence $\operatorname{sgn}\det[F_r^{(p+q)}(0)] = (-1)^{\binom{k}{2}}$, and Corollary 3.2 gives the claim. ∎

Proof of Theorem 1.1(i). By Proposition 5.2, every $\Delta_k^{(r)}$ is nonzero, so by Section 2 every leading Gram matrix is nondegenerate and the polynomials P_n exist. Let $n = r+dj$. Replacing z^n by P_n in the basis $z^r, z^{r+d},\dots,z^{r+dj}$ is a unitriangular change of basis, and orthogonality makes the new Gram matrix $\operatorname{diag}\big([G_{pq}^{(r)}]_{p,q=0}^{j-1},(P_n,P_n)_a\big)$. Comparing determinants gives

$$(P_n,P_n)_a = \frac{\Delta_{j+1}^{(r)}}{\Delta_j^{(r)}}; \tag{5.1}$$

this Schur-complement identity uses only the nonvanishing of the leading minors, not positive definiteness. By Proposition 5.2 and $\binom{j+1}{2} - \binom{j}{2} = j$, $\operatorname{sgn}(P_n,P_n)_a = (-1)^{r+j}$, which is $\varrho(n)$ by (2.7). Hence $\mathfrak{h}_n = \varrho(n)(P_n,P_n)_a > 0$. The last assertion is the identification at the end of Section 2. ∎

Remark 5.3. Unwinding the proof of Lemma 5.1 writes $(-1)^{rk+\binom{k}{2}}\Delta_k^{(r)}$ as a $k(d-1)$-fold integral of a positive function. By (5.1), the norms $\mathfrak{h}_n$ and the ratios V_n can therefore be evaluated without the cancellation inherent in the oscillatory moment determinants.

6. The recurrence

Let $d \geq 3$. Define $V_n = \mathfrak{h}_{n+1}/\mathfrak{h}_n$ for $n \geq 0$ and $V_j = 0$ for $j < 0$. By Theorem 1.1(i), $V_n > 0$ for every $n \geq 0$. We derive (1.1), following [FH25, Lemma 3.4 and equations (3.1)–(3.3)].

If g has weight ℓ, then

$$\sigma(zg) \;=\; -\tau(\ell) z\,\sigma g. \tag{6.1}$$

For $\ell \leq d-2$ this follows from $(-\eta)^{-\ell-1}\eta = -(-\eta)^{-\ell}$; for $\ell = d-1$ the weight returns to 0, and $\sigma(zg) = \eta z\, g(\eta z) = \eta(-\eta)^{d-1} z\,\sigma g = (-1)^{d-1}\eta^d z\,\sigma g = -\tau(d-1) z\,\sigma g$.

Let $L = \partial_z + \mathrm{i}z^{d-1}$, where $\partial_z = \frac{1}{2}(\partial_{\operatorname{Re} z} - \mathrm{i}\partial_{\operatorname{Im} z})$. Since $\partial_z \overline{\sigma g(z)} = 0$ and $\partial_z \rho_a(z) = (-a\bar{z} + \mathrm{i}z^{d-1})\rho_a(z)$, integration by parts in (2.2), with no boundary terms because of the Gaussian decay, and (6.1) give, for g of weight ℓ,

$$(Lf, g)_a \;=\; a\int_{\mathbb{C}} f\,\bar{z}\,\overline{\sigma g}\,\rho_a\,\mathrm{d}A \;=\; -\tau(\ell) a (f, zg)_a. \tag{6.2}$$

The polynomials zP_n and LP_n have weights $n+1$ and $n-1$ modulo d, degrees $n+1$ and $n+d-1$, and leading coefficients 1 and i. If $j \leq n+1-2d$ and $j \equiv n+1 \pmod d$, then $\deg LP_j < n$, so (6.2) gives $0 = (LP_j, P_n)_a = -\tau(n) a (P_j, zP_n)_a$. If $j \leq n-1-d$ and $j \equiv n-1 \pmod d$, then $\deg zP_j < n$, so $(LP_n, P_j)_a = -\tau(j) a (P_n, zP_j)_a = 0$. Hence

$$zP_n \;=\; P_{n+1} + c_n P_{n+1-d}, \qquad LP_n \;=\; \mathrm{i}P_{n+d-1} + d_n P_{n-1}, \tag{6.3}$$

where terms with negative indices are omitted; equivalently, $c_n = 0$ for $n \leq d-2$, and c_n for $n \geq d-1$ and d_n for $n \geq 1$ are the coefficients displayed. The $d-1$ consecutive integers $n-d+1, \dots, n-1$ miss exactly the residue of n modulo d, and the product of τ over a full residue system is $(-1)^{d-1}$; hence (2.6) gives

$$\varrho(n) \;=\; \tau(n)\varrho(n-d+1) \quad (n \geq d-1), \qquad \varrho(n) \;=\; -\tau(n-1)\varrho(n-1) \quad (n \geq 1).$$

Pairing (6.3) with P_{n+1-d} (for $n \geq d-1$) and P_{n-1} (for $n \geq 1$), and using (6.2) and Hermitian symmetry, we obtain

$$\begin{aligned}
\mathrm{i}\varrho(n)\mathfrak{h}_n \;&=\; (LP_{n+1-d}, P_n)_a \;=\; -\tau(n) a (P_{n+1-d}, zP_n)_a, \\
c_n \;&=\; \frac{(zP_n, P_{n+1-d})_a}{\varrho(n+1-d)\mathfrak{h}_{n+1-d}} \;=\; \frac{\mathrm{i}}{a}\frac{\mathfrak{h}_n}{\mathfrak{h}_{n+1-d}}, \\
d_n \;&=\; \frac{(LP_n, P_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} \;=\; -\tau(n-1) a \frac{(P_n, zP_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} \;=\; a\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}};
\end{aligned}$$

the first line holds because the leading term of LP_{n+1-d} is $\mathrm{i}P_n$, and the last because that of zP_{n-1} is P_n. Thus $c_m = \frac{\mathrm{i}}{a}\prod_{i=m+1-d}^{m-1} V_i$ for every $m \geq 0$, including $c_m = 0$ for $m \leq d-2$.

Applying the first relation in (6.3) $d-1$ times to P_n, a path from P_n to P_{n-1} in $d-1$ steps of $+1$ or $-(d-1)$ uses exactly one step of $-(d-1)$. So the coefficient of P_{n-1} in $z^{d-1}P_n$ is $\sum_{s=0}^{d-2} c_{n+s}$. Let $n \geq 1$. The coefficient of P_{n-1} in $\partial_z P_n$ is n, since $\partial_z P_n = nz^{n-1} + \cdots$ lies in the weight space of P_{n-1}. Comparing coefficients of P_{n-1} in $LP_n = \partial_z P_n + \mathrm{i}z^{d-1}P_n$ and using $d_n = aV_{n-1}$, we obtain, with $m = n-1 \geq 0$,

$$aV_m \;=\; m+1-\frac{1}{a}\sum_{s=0}^{d-2}\;\prod_{i=m+s-d+2}^{m+s} V_i,$$

which is (1.1).

Proof of Theorem 1.1(ii). The sequence $V_n = \mathfrak{h}_{n+1}/\mathfrak{h}_n$ is positive and satisfies (1.1). For $j \leq d-1$, $P_j = z^j$, $\sigma z^j = (-1)^j z^j$ by (2.3), and $\varrho(j) = (-1)^j$; hence $\mathfrak{h}_j = \int_{\mathbb{C}} |z|^{2j}\rho_a \,\mathrm{d}A$, and V_j is given by (1.2) for $j \leq d-3$. If $(\widetilde{V}_n)$ is another real solution with these initial values and $\widetilde{V}_i = V_i$ for $i \leq m+d-3$, then (1.1) at index m is linear in $\widetilde{V}_{m+d-2}$ with coefficient $a^{-2}\widetilde{V}_m \cdots \widetilde{V}_{m+d-3} = a^{-2}V_m \cdots V_{m+d-3} > 0$, so $\widetilde{V}_{m+d-2} = V_{m+d-2}$. Induction proves uniqueness. ∎

7. The quartic case

In this section $d = 4$. Let $y > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $b = \beta + \frac{1}{2}$. For $w > 0$, define

$$\Phi_{\alpha,\beta}(w;y) \;=\; \int_0^\infty \mathrm{e}^{-ys}s^{\alpha-1/2}(1+ws^2)^{-b}\,\mathrm{d}s. \tag{7.1}$$

Throughout, $y = a^2/2$ and $C = \sqrt{\pi/2}$.

Lemma 7.1 *(quartic kernels).* *Define*

r	A_r	α_r	β_r
0	C	0	0
1	aC	1	0
2	$2C$	1	1

(7.2)

Then $\mathcal{W}_r(a;\lambda) = A_r\Phi_{\alpha_r,\beta_r}(\lambda^2;y)$ *for* $r = 0,1,2$ *and* $\lambda > 0$.

Proof. Let $\nu = \alpha + \frac{1}{2}$. Since $(1+ws^2)^{-b} = \mathbb{E}[\mathrm{e}^{-ws^2G_b}]$ for a Gamma variable G_b with shape b, and $\mathrm{e}^{-ys}s^{\nu-1}$ is $\Gamma(\nu)y^{-\nu}$ times the density of G_ν/y,

$$\Phi_{\alpha,\beta}(w;y) \;=\; \Gamma(\nu)y^{-\nu}\,\mathbb{E}\left[\exp\bigl(-wy^{-2}G_\nu^2G_b\bigr)\right]$$

with G_ν and G_b independent. By the duplication formula [OLBC10, equation (5.5.5)], G_ν^2 has the same Mellin transform, and hence the same law, as $4G_{\nu/2}G_{(\nu+1)/2}$ with independent factors. So $\Phi_{\alpha,\beta}(w;y) = \Gamma(\nu)y^{-\nu}\mathbb{E}[\exp(-4wy^{-2}G_{\nu/2}G_{(\nu+1)/2}G_b)]$. For $(\alpha,\beta) = (\alpha_r,\beta_r)$ the shapes $\{\nu/2,(\nu+1)/2,b\}$ are $\{\frac14,\frac34,\frac12\}$, $\{\frac34,\frac54,\frac12\}$ and $\{\frac34,\frac54,\frac32\}$, which are the shapes $\beta_{r,j}$, $j \in J_r$, of Section 4 for $r = 0,1,2$. Moreover $4y^{-2} = 16a^{-4} = d^{\,d-2}a^{-d}$ and $A_r\Gamma(\nu)y^{-\nu} = \pi r!\,a^{-r-1}$. The claim follows from Proposition 4.1. ∎

Lemma 7.2 *(Bessel form).* *For every $a > 0$, $\int_{\mathbb{C}} \rho_a \, \mathrm{d}A = h(a)$, the Bessel combination (1.5).*

Proof. Put $H(a) = \int_{\mathbb{C}} \rho_a \, \mathrm{d}A$ and $m_{pq} = \int_{\mathbb{C}} z^p \bar{z}^q \rho_a(z) \, \mathrm{d}A(z)$ for $a > 0$.

Step 1: $H''' + a^2 H' + aH = 0$ on $(0, \infty)$. Differentiation under the integral sign gives $H^{(k)}(a) = (-1)^k m_{kk}$. With the Wirtinger derivative $\partial_z = \frac{1}{2}(\partial_{\mathrm{Re}\, z} - \mathrm{i}\partial_{\mathrm{Im}\, z})$, $\partial_z \bar{z}^q = 0$ and $\partial_z \rho_a = (-a\bar{z} + \mathrm{i}z^3)\rho_a$, so integrating $\partial_z(z^p \bar{z}^q \rho_a)$ over $\mathbb{C}$, with no boundary terms because of the Gaussian decay, gives

$$p\, m_{p-1,q} + \mathrm{i}\, m_{p+3,q} = a\, m_{p,q+1}.$$

For $(p, q) = (1, 0)$ and $(0, 3)$ this reads $m_{00} + \mathrm{i}m_{40} = am_{11}$ and $\mathrm{i}m_{33} = am_{04}$. The substitution $z \mapsto \eta z$ of Section 2 gives $\overline{m_{qp}} = \bar{\eta}^p \eta^q m_{pq}$; hence every m_{kk} is real and $\overline{m_{40}} = -m_{04}$. Therefore $m_{33} = am_{00} - a^2 m_{11}$, which is the claimed equation.

Step 2: the Bessel side. For $x > 0$, $J_\nu(x) = (x/2)^\nu s_\nu(x)$ with the even entire function $s_\nu(x) = \sum_{k \geq 0} (-1)^k x^{2k} / \big(4^k k! \, \Gamma(k + \nu + 1)\big)$ [OLBC10, equation (10.2.2)], and Bessel's equation [OLBC10, equation (10.2.1)] becomes $x s_\nu'' + (2\nu + 1) s_\nu' + x s_\nu = 0$. Writing $s_\pm = s_{\pm 1/4}$ and $x = a^2/4$, so that $(x/2)^{\mp 1/4} = (a^2/8)^{\mp 1/4}$, we get, for $a > 0$,

$$h(a) = h_S(a) := \frac{\pi^2}{4} \left[\sqrt{8}\, s_-(x)^2 - \sqrt{2}\, a\, s_-(x) s_+(x) + \frac{a^2}{\sqrt{8}}\, s_+(x)^2 \right].$$

The function h_S is smooth on $\mathbb{R}$. Put $U(a) = s_-(a^2/4)$ and $V(a) = a\, s_+(a^2/4)$. The equations for $s_\mp$, with $2\nu + 1 = \frac{1}{2}$ and $\frac{3}{2}$, give $U'' + \frac{a^2}{4} U = 0$ and $V'' + \frac{a^2}{4} V = 0$. If y_1, y_2 solve $y'' + qy = 0$, then $P = y_1 y_2$ satisfies $P''' + 4qP' + 2q'P = 0$; with $q = a^2/4$ this is $P''' + a^2 P' + aP = 0$. Since $h_S = \frac{\pi^2}{4}\big(\sqrt{8}\, U^2 - \sqrt{2}\, UV + V^2/\sqrt{8}\big)$, we get $h_S''' + a^2 h_S' + a h_S = 0$ on $\mathbb{R}$. Since $s_\nu(0) = 1/\Gamma(\nu + 1)$ and $s_\nu'(0) = 0$,

$$h_S(0) = \frac{\pi^2 \sqrt{8}}{4\, \Gamma(\frac{3}{4})^2}, \qquad h_S'(0) = -\frac{\pi^2 \sqrt{2}}{4\, \Gamma(\frac{3}{4}) \Gamma(\frac{5}{4})}, \qquad h_S''(0) = \frac{\pi^2}{2\sqrt{8}\, \Gamma(\frac{5}{4})^2}.$$

Step 3: the planar side as $a \to 0^+$. By Lemma 7.1 with $\lambda = 1$,

$$\begin{aligned} H &= C\, \Phi_{0,0}(1; y), \\ H' &= -\mathcal{W}_1(a; 1) = -aC\, \Phi_{1,0}(1; y), \\ H'' &= \mathcal{W}_2(a; 1) = 2C\, \Phi_{1,1}(1; y). \end{aligned}$$

As $y \downarrow 0$, monotone convergence and $\int_0^\infty s^{2x-1}(1 + s^2)^{-x-z} \, \mathrm{d}s = \frac{1}{2} B(x, z) = \frac{\Gamma(x)\Gamma(z)}{2\Gamma(x+z)}$ [OLBC10, equations (5.12.1) and (5.12.3)] give $\Phi_{0,0}(1; y) \to \frac{1}{2} B(\frac{1}{4}, \frac{1}{4}) = \Gamma(\frac{1}{4})^2/(2\sqrt{\pi})$ and $\Phi_{1,1}(1; y) \to \frac{1}{2} B(\frac{3}{4}, \frac{3}{4}) = \Gamma(\frac{3}{4})^2/\sqrt{\pi}$. The substitution $s = u/y$ gives $a\Phi_{1,0}(1; y) = \sqrt{2} \int_0^\infty \mathrm{e}^{-u} u^{1/2} (y^2 + u^2)^{-1/2} \, \mathrm{d}u$; the integrand is bounded by $\mathrm{e}^{-u} u^{-1/2}$, so by dominated convergence this tends to $\sqrt{2}\, \Gamma(\frac{1}{2}) = \sqrt{2\pi}$. Using $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$, $\Gamma(\frac{5}{4}) = \frac{1}{4}\Gamma(\frac{1}{4})$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ [OLBC10, equations (5.5.3), (5.5.1), (5.4.6)], the limits of H, H', H'' as $a \to 0^+$ are $h_S(0)$, $h_S'(0) = -\pi$ and $h_S''(0)$.

Step 4: uniqueness. The function $W = H - h_S$ is smooth on $(0, \infty)$, satisfies $W''' = -a^2 W' - aW$, and $W, W', W'' \to 0$ as $a \to 0^+$. Fix $a_1 > 0$. For $0 < a \leq a_1$, the energy $E = W^2 + W'^2 + W''^2$ satisfies $E' \leq (2 + a_1^2 + a_1) E$, by $2|uv| \leq u^2 + v^2$. Hence

$\mathrm{e}^{-(2+a_1^2+a_1)a}E(a)$ is nonincreasing on $(0,a_1]$ and tends to 0 as $a\to 0^+$, so $E=0$ there. Thus $H=h_S=h$ on $(0,\infty)$. ∎

The identity of Lemma 7.2 also follows from Lemma 7.1 and the Dixon–Ferrar formula [GR15, entry 6.518]: substitute $s=\sinh t$ in $\Phi_{0,0}(1;y)$, use $K_{1/2}(z)=\sqrt{\pi/(2z)}\,\mathrm{e}^{-z}$, take $\nu=\frac14$ and $x=a^2/4$ in the formula, and use $Y_{1/4}=J_{1/4}-\sqrt{2}\,J_{-1/4}$.

Proof of Theorem 1.2. With $d=4$, $V_m=av_m$ and $\epsilon=a^{-2}$, (1.1) is (1.3). By Lemma 7.2, $h(a)=\int_{\mathbb{C}}\rho_a\,\mathrm{d}A$, and differentiation under this absolutely convergent integral gives $h'(a)=-\int_{\mathbb{C}}|z|^2\rho_a\,\mathrm{d}A$ and $h''(a)=\int_{\mathbb{C}}|z|^4\rho_a\,\mathrm{d}A$. By the proof of Theorem 1.1(ii),

$$\mathfrak{h}_0=h,\qquad \mathfrak{h}_1=-h',\qquad \mathfrak{h}_2=h''.$$

Hence $h(a)>0$ and $h'(a)<0$, so (1.4) is well defined, and $v_n=\mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$ reproduces it. Theorem 1.1(ii) for $d=4$ now gives existence, uniqueness and positivity. (For $d=4$ the initial data of (1.2) consist of V_0 and V_1, that is, of v_0 and v_1.) ∎

8. Uniqueness of the positive solution

Proof of Theorem 1.3. Fix $\epsilon>0$ and let (v_n) and (v_n') be positive solutions of (1.3). Put $T_n=v_nv_{n+1}v_{n+2}$ and $T_{-1}=T_{-2}=0$, and similarly T_n'. Since $v_n(v_{n-1}v_{n-2}+v_{n+1}v_{n-1}+v_{n+2}v_{n+1})=T_{n-2}+T_{n-1}+T_n$, the recurrence reads

$$v_n+T_n+T_{n-1}+T_{n-2}=(n+1)\epsilon,\qquad n\ge 0. \tag{8.1}$$

Step 1: a priori bounds. All terms in (8.1) are nonnegative, so $v_n\le(n+1)\epsilon$ and $T_n\le(n+1)\epsilon$. Hence, bounding each factor of the bracket in (1.3) by these estimates (for $n=0,1$ the products containing a negative index vanish), the bracket is at most $1+\epsilon^2[n(n-1)+n(n+2)+(n+2)(n+3)]=1+3\epsilon^2((n+1)^2+1)\le K(n+1)^2$ with $K=1+6\epsilon^2$, and therefore

$$v_n\ge\frac{c}{n+1},\qquad c=\frac{\epsilon}{K}. \tag{8.2}$$

The same bounds hold for v'. With $\delta_n=\log(v_n'/v_n)$, it follows that $|\delta_n|\le\log K+2\log(n+1)$.

Step 2: exact linearization. For $x,y>0$, the logarithmic mean is $L(x,y)=(y-x)/(\log y-\log x)$ if $x\ne y$ and $L(x,x)=x$; equivalently $L(x,y)=\int_0^1 x^{1-t}y^t\,\mathrm{d}t$. Let $a_n=L(v_n,v_n')$ and $b_n=L(T_n,T_n')$, so that $v_n'-v_n=a_n\delta_n$ and $T_n'-T_n=b_nD_n$ with $D_n=\delta_n+\delta_{n+1}+\delta_{n+2}$. A logarithmic mean lies between its arguments, so by Step 1

$$a_n\ge\frac{c}{n+1},\qquad 0<b_n\le(n+1)\epsilon. \tag{8.3}$$

Subtracting (8.1) for the two solutions gives

$$a_n\delta_n+b_nD_n+b_{n-1}D_{n-1}+b_{n-2}D_{n-2}=0,\qquad n\ge 0, \tag{8.4}$$

where terms with negative indices are omitted.

Step 3: a localized energy identity. Let $(h_n)_{n\ge 0}$ be real with finite support. Multiplying (8.4) by $h_n^2\delta_n$, summing over n, and collecting the terms with the same b_m gives

$$\sum_{n\ge 0} a_n h_n^2\delta_n^2 + \sum_{m\ge 0} b_m\Big(\sum_{r=0}^{2}\delta_{m+r}\Big)\Big(\sum_{r=0}^{2} h_{m+r}^2\delta_{m+r}\Big) \;=\; 0.$$

For real z_0,z_1,z_2 and h_0,h_1,h_2,

$$\Big(\sum_r z_r\Big)\Big(\sum_r h_r^2 z_r\Big) \;=\; \Big(\sum_r h_r z_r\Big)^2 + \sum_{0\le r<s\le 2}(h_r-h_s)^2 z_r z_s.$$

Dropping the nonnegative squares, using $2|z_rz_s|\le z_r^2+z_s^2$ and (8.3), we obtain

$$c\sum_{n\ge 0}\frac{h_n^2\delta_n^2}{n+1} \;\le\; \frac{\epsilon}{2}\sum_{m\ge 0}(m+1)\sum_{0\le r<s\le 2}(h_{m+r}-h_{m+s})^2\big(\delta_{m+r}^2+\delta_{m+s}^2\big). \tag{8.5}$$

Step 4: a power weight. Let $0<p<1$, put $w_n=(n+1)^{-p}$, and

$$Q_p \;=\; \sum_{n\ge 0}\frac{\delta_n^2}{(n+1)^{1+2p}},$$

which is finite by the logarithmic bound of Step 1. For an integer $R\ge 1$, the sequence $h_n=(w_n-w_R)_+$ is supported on $n<R$, and $|h_i-h_j|\le|w_i-w_j|$. By the mean value theorem, $|w_{m+r}-w_{m+s}|\le p(s-r)(m+1)^{-p-1}$. In (8.5) the three pairs (r,s) contribute the squared distances 1, 4, 1, so the right-hand side is at most

$$\frac{\epsilon p^2}{2}\sum_{m\ge 0}\frac{5\delta_m^2+2\delta_{m+1}^2+5\delta_{m+2}^2}{(m+1)^{1+2p}} \;\le\; \frac{\epsilon p^2}{2}\cdot 12\cdot 27\,Q_p \;=\; 162\,\epsilon p^2 Q_p,$$

where we used $(m+r+1)/(m+1)\le 3$ and $1+2p\le 3$. As $R\to\infty$, $h_n\uparrow w_n$, and monotone convergence on the left of (8.5) gives $cQ_p\le 162\,\epsilon p^2 Q_p$. Choosing $p=1/(18\sqrt{K})$ makes $162\,\epsilon p^2=c/2$, so $cQ_p\le\frac{c}{2}Q_p$. Since $Q_p<\infty$, $Q_p=0$, hence $\delta_n=0$ and $v_n'=v_n$ for every n. ∎

With a_n,b_n held fixed, (8.4) is the Euler–Lagrange equation of the quadratic form $\sum_n a_n\delta_n^2+\sum_n b_n(\delta_n+\delta_{n+1}+\delta_{n+2})^2$. In the variables $u_n=\log v_n$, (8.1) itself is the coordinatewise stationarity condition of the formal energy $\sum_n\big[\mathrm{e}^{u_n}+\mathrm{e}^{u_n+u_{n+1}+u_{n+2}}-(n+1)\epsilon\,u_n\big]$, whose increments along finitely supported variations are well defined and strictly convex. Convexity alone does not compare two infinite sequences; the power-weighted estimate of Steps 3 and 4 controls the boundary at infinity.

Remark 8.1. For general $d\ge 3$, the substitution $V_n=a^{2/(d-2)}v_n$ and $\epsilon=a^{-d/(d-2)}$ turns (1.1) into $v_n+\sum_{m=n-d+2}^{n}T_m=(n+1)\epsilon$, where T_m is the product of the $d-1$ consecutive terms $v_m,\dots,v_{m+d-2}$. The upper bounds of Step 1 persist, but the same argument then gives only $v_n\ge c(n+1)^{-(d-3)}$. For $d\ge 5$ this is too weak for Step 4. Using only $a_n\ge c(n+1)^{-A}$ and $b_n\le(n+1)\epsilon$, Step 4 needs a weight $w_n>0$ decreasing to 0 with $\epsilon(w_n-w_{n+1})^2\lesssim c(n+1)^{-A-1}w_n^2$. When $A>1$ the relative decrements $(w_n-w_{n+1})/w_n$ are then summable, so w_n cannot tend to 0. Thus the present estimates do not extend Theorem 1.3 to $d\ge 5$.

AI-use disclosure

The proof of the case $d = 4$ of Theorem 1.1, in an earlier quartic-specific form, was developed with assistance from OpenAI Codex using GPT-5.6 Sol (`gpt-5.6-sol`) at high and extra-high reasoning effort. The extension to every d (the Gamma representation and the total-positivity argument of Sections 4 and 5) was developed with assistance from Anthropic's Claude Code using Claude Opus 5.5. The manuscript was revised, and the Lean formalization of the case $d = 4$ was produced, with assistance from Claude Code using Claude Opus 5.5, Claude Fable 5 and Claude Haiku 5.5. The weighted energy estimate in the proof of Theorem 1.3 was obtained in a consultation with OpenAI's GPT-6 Pro, which was also used for independent referee-style checks of the proofs, the citations and the Lean statements. These systems were used as interactive research assistants to develop and stress-test the arguments, audit citations and calculations, write the formal proofs, and revise the manuscript. The author directed the research, selected and checked the final arguments, and takes responsibility for all mathematical claims, citations, computations, and wording.

Code availability

The Lean 4 formalization of the case $d = 4$ of Theorem 1.1 and of Theorems 1.2 and 1.3 is in the directory `lean` of release `v2.0` of the repository github.com/skypher/painleve (Lean 4.33.0, Mathlib commit `db584cd6d46c`). It follows the quartic argument of the version of this paper in release `v1.1`. The statements `Painleve.Stmt_main` (the case $d = 4$ of Theorem 1.1 with Theorem 1.2) and `Painleve.Stmt_unique` (Theorem 1.3) are proved by `Painleve.main_theorem` and `Painleve.unique_theorem`. These proofs use only the axioms `propext`, `Classical.choice` and `Quot.sound`; the repository README gives the build commands and the file `lean/Check.lean` that checks both. The directory `runs/general-d` contains numerical checks of Sections 3–6 for $d \leq 8$, and `runs/uniq` the numerical scans of initial values used as a sanity check for Theorem 1.3.

References

[AHK19] J. Arnlind, J. Hoppe and M. Kontsevich, *Quantum minimal surfaces*, arXiv:1903.10792 (2019).

[And86] C. Andréief, *Note sur une relation entre les intégrales définies des produits des fonctions*, Mém. Soc. Sci. Phys. Nat. Bordeaux (3) **2** (1886) 1–14.

[Bor99] A. Borodin, *Biorthogonal ensembles*, Nucl. Phys. B **536** (1999) 704–732, doi:10.1016/S0550-3213(98)00642-7.

[CDHM25] P. A. Clarkson, A. V. Dzhamay, A. N. W. Hone and B. Mitchell, *Special solutions of a discrete Painlevé equation for quantum minimal surfaces*, Theor. Math. Phys. **224** (2025) 1359–1397, doi:10.1134/S0040577925080045, arXiv:2503.14436.

[CT99] L. Cornalba and W. Taylor IV, *Holomorphic curves from matrices*, Nucl. Phys. B **536** (1999) 513–552, doi:10.1016/S0550-3213(98)00702-0.

[FH25] G. Felder and J. Hoppe, *Orthogonal polynomials with complex densities and quantum minimal surfaces*, SIGMA **21** (2025) 103, doi:10.3842/SIGMA.2025.103, arXiv:2504.06197.

[For19] P. J. Forrester, *Meet Andréief, Bordeaux 1886, and Andreev, Kharkov 1882–1883*, Random Matrices Theory Appl. **8** (2019) 1930001, doi:10.1142/S2010326319300018, arXiv:1806.10411.

[GR15] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*, 8th ed., edited by D. Zwillinger and V. H. Moll, Academic Press, Amsterdam (2015).

[Hop25] J. Hoppe, *Quantum states from minimal surfaces*, arXiv:2502.18422 (2025).

[OLBC10] F. W. J. Olver, D. W. Lozier, R. F. Boisvert and C. W. Clark (eds.), *NIST Handbook of Mathematical Functions*, Cambridge University Press, New York (2010), dlmf.nist.gov.